KidWind 2: Change one thing

Goal: I can change one thing about my blades and keep everything else the same.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 11, grade 5 path; Playbook B meeting 12, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Tell the class which set lifted the cup higher and by how many centimeters.

You will use

- One teacher-run test station, built and piloted first from parts on hand
- One guarded fan at one fixed setting, with tape marks and a ruler
- A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub
- String on a fixed spool, one small cup and the same load pieces every trial
- Cardstock or index cards, reusable blade supports, tape and student scissors
- A marked blade-angle example, timer, shallow catch tray, team folders and record sheets
- Printed practice data table for the no-fan paper path

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Find your team folder and your Set A rotor.
- Read the goal: change one thing only.

2 Step 2: Read the chart — 1 min

- Look at last time's class chart.

3 Step 3: Watch a fair test — 4 min

- Watch the teacher change only the blade angle.
- Notice what stays the same.

4 Step 4: Build Set B — 12 min

- Pick one thing to change: number, size, angle or shape.
- Keep everything else the same as Set A.
- Write the one change on your record sheet.

5 Step 5: Test A and B — 11 min

- Test Set A once.
- Test Set B once.
- Record centimeters lifted for each trial.

6 Step 6: Reset the station — 3 min

- Fan off first.
- Rotor sets go flat in your team folder.
- Cup and load pieces go back in the station tray.

7 Step 7: Share — 2 min

- Tell the class which set lifted the cup higher and by how many centimeters.

What to turn in

Set A and Set B results in centimeters, with the one change named. These are first results.

No computer? Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Finished early? Next challenge

Predict which change would help most next time and write why.

Stay safe

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.

No-fan paper path

Use these paper steps instead when your teacher chooses the no-fan path.

Goal: I can explain how wind can lift a load and use sample results to plan a fair test.

1. Compare design A with design B.
2. Name the one stated change.
3. List what the test must keep the same.
4. Read all three trials for each.

Keep: A one-change sentence and a fair-test checklist.

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Unit: Centimeters lifted in 10 seconds with one unchanged cup and load

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15